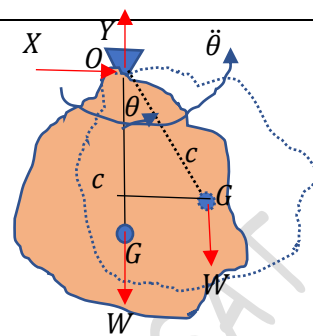


Compound Pendulum

Any rigid body suspended in such a way that it is free to rotate about any fixed horizontal axis and perpendicular to the plane is known as compound pendulum.

Consider a rigid body of any arbitrary shape suspended at O . It can rotate freely about the perpendicular axis passing through O . When the body is in static equilibrium, the centre of gravity G will be directly under the point of suspension O . Let the centre of gravity be at a distance c from the point of suspension. Let the body be disturbed from its static equilibrium position. Then the body will execute small rotational oscillations about the axis through O .



Consider an instant when the body rotates through an angle θ in the counter clockwise direction with an angular acceleration $\ddot{\theta}$. The equation of motion of the system can be written as,

$$I_O \ddot{\theta} = \Sigma M_O$$

$$I_O \ddot{\theta} = -W c \sin \theta$$

For small angles of rotation, $\sin \theta \cong \theta$. Hence the equation of motion can be written as

$$\ddot{\theta} + \frac{Wc}{I_O} \theta = 0$$

This is now of the form $\ddot{\theta} + p^2 \theta = 0$, where $p = \sqrt{\frac{Wc}{I_O}}$.

The period of small oscillations $T = \frac{2\pi}{p} = 2\pi \sqrt{\frac{I_O}{Wc}}$

If i_o is the radius of gyration of the body with respect to the axis through O , then $I_O = \frac{W}{g} i_o^2$

$$T = 2\pi \sqrt{\frac{\frac{W}{g} i_o^2}{Wc}} = 2\pi \sqrt{\frac{i_o^2/c}{g}}$$

If we compare this equation with the equation for the period of oscillations of a simple pendulum ($= 2\pi \sqrt{\frac{L}{g}}$), we can write the period of oscillation of a compound pendulum as,

$$T = 2\pi \sqrt{\frac{L_e}{g}}$$

Where L_e is known as the equivalent length of the compound pendulum

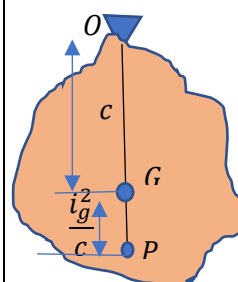
$$L_e = i_o^2/c$$

By parallel axis theorem, we can write $I_O = I_G + \frac{W}{g} c^2$

$$\frac{W}{g} i_o^2 = \frac{W}{g} i_g^2 + \frac{W}{g} c^2 \gggggggg i_o^2 = i_g^2 + c^2$$

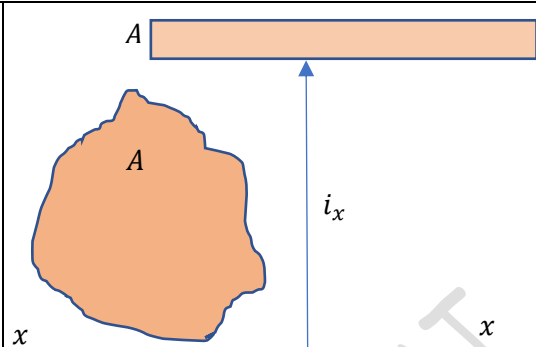
$$L_e = \frac{i_o^2}{c} = \frac{i_g^2 + c^2}{c} = c + \frac{i_g^2}{c}$$

The point P is known as centre of percussion or centre of oscillation.



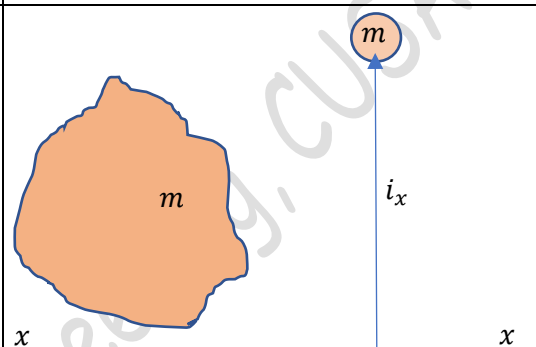
Radius of gyration

Consider a plane area A . Let it be having a moment of inertia I_{xx} with respect to any $x-x$ axis. Let us imagine that this area is squeezed in to a thin strip parallel to the $x-x$ axis. The distance at which this strip of area A be placed from the $x-x$ axis so that it will be having the same moment of inertia I_{xx} is known as the radius of gyration i_x of the area with respect to the $x-x$ axis. By definition, $I_{xx} = A i_x^2$



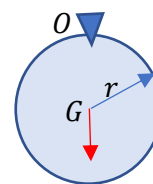
Similarly, consider a body of mass m . The mass moment of inertia of the body with respect to an axis $x-x$. Let it be having a mass moment of inertia I_{xx} . Let this body be squeezed in to a ball without any loss in mass. The distance i_x at which this ball be placed so that it will have the same moment of inertia I_{xx} as the original body is known as the radius of gyration.

$$I_{xx} = m i_x^2$$



Example 1

A homogeneous circular disc of radius r and weight W hangs in a vertical plane from a pin at O on its circumference. Find the period of small rotational oscillations of the disc.



The period of oscillations of a compound pendulum is given by the equation, $T = 2\pi \sqrt{\frac{L_e}{g}}$

The equivalent length of the compound pendulum, $L_e = c + \frac{i_g^2}{c}$

Distance of the centre of gravity from the axis of rotation, $c = r$

Mass moment of inertia of the disc with respect to the perpendicular axis passing through G

$$I_{GG} = \frac{W}{g} \frac{r^2}{2}$$

If i_g is the radius of gyration of the disc with respect to the perpendicular axis through G

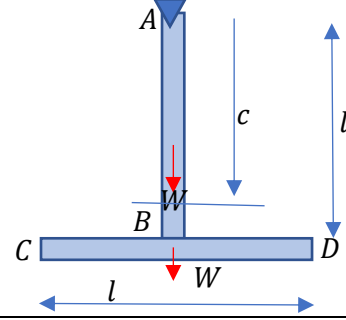
$$I_{GG} = \frac{W}{g} i_g^2 \ggggggg \text{ i.e., } \frac{W}{g} i_g^2 = \frac{W}{g} \frac{r^2}{2} \gggg i_g^2 = \frac{r^2}{2}$$

$$\therefore L_e = c + \frac{i_g^2}{c} = r + \frac{r^2}{2r} = \frac{3r}{2}$$

$$T = 2\pi \sqrt{\frac{L_e}{g}} = 2\pi \sqrt{\frac{\frac{3r}{2}}{g}} = 2\pi \sqrt{\frac{3r}{2g}}$$

Example 2

Find the period of small oscillations of the compound pendulum shown. Bars AB and CD are of equal length, identical weight, and uniform cross section.



The period of oscillations of a compound pendulum is given by the equation, $T = 2\pi \sqrt{\frac{L_e}{g}}$

The equivalent length of the compound pendulum, $L_e = \frac{i_A^2}{c} = c + \frac{i_g^2}{c}$, where c is the distance of centre of gravity of the pendulum from the point of suspension.

Let each bar has a length l and weight W . The mass moment of inertia of the compound pendulum with respect to the perpendicular axis through A.

$$I_A = (I_{AB})_A + (I_{CD})_A$$

$$I_{AB} = \frac{W}{g} \frac{l^2}{12} + \frac{W}{g} \left(\frac{l}{2}\right)^2 = \frac{Wl^2}{3g} \quad \lll \ggg \quad I_{CD} = \frac{W}{g} \frac{l^2}{12} + \frac{W}{g} (l)^2 = \frac{13Wl^2}{12g}$$

$$I_A = I_{AB} + I_{CD} = \frac{Wl^2}{3g} + \frac{13Wl^2}{12g} = \frac{17Wl^2}{12g}$$

If i_A is the radius of gyration of the pendulum with respect to the perpendicular axis through A, we can write, $I_A = \frac{2W}{g} i_A^2$ where $2W$ is the total weight of the pendulum.

$$i.e., \quad \frac{2W}{g} i_A^2 = \frac{17Wl^2}{12g} \quad \ggg \quad i_A^2 = \frac{17l^2}{24}$$

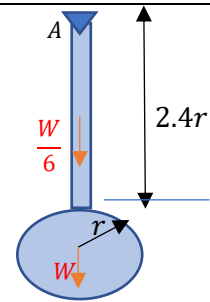
The distance of c.g from the axis of rotation, $c = \frac{W \times \frac{l}{2} + W \times l}{W + W} = \frac{3}{4}l$

$$L_e = \frac{i_A^2}{c} = \frac{\frac{17l^2}{24}}{\frac{3}{4}l} = \frac{17 \times 4l}{3 \times 24} = \frac{17l}{18}$$

$$T = 2\pi \sqrt{\frac{L_e}{g}} = 2\pi \sqrt{\frac{17l}{18g}} = 6.11 \sqrt{\frac{l}{g}}$$

Example 3

Find the period of small oscillations of the compound pendulum shown which consists of a disc of weight W and radius r suspended by a slender rod of weight $\frac{W}{6}$ and length $2.4r$.



The period of oscillations of a compound pendulum is given by the equation, $T = 2\pi \sqrt{\frac{I_e}{g}}$

The equivalent length of the compound pendulum, $L_e = \frac{i_A^2}{c} = c + \frac{i_g^2}{c}$, where c is the distance of centre of gravity of the pendulum from the point of suspension.

The mass moment of inertia of the compound pendulum with respect to the perpendicular axis through A.

$$I_A = (I_{rod})_A + (I_{disc})_A$$

$$(I_{rod})_A = \frac{\frac{W}{6}}{g} \frac{(2.4r)^2}{12} + \frac{\frac{W}{6}}{g} \left(\frac{2.4r}{2}\right)^2 = \frac{0.32Wr^2}{g} \quad \ll\ll\ll\ll\ll \quad (I_{disc})_A = \frac{W}{g} \frac{r^2}{2} + \frac{W}{g} (3.4r)^2 = \frac{12.06Wr^2}{g}$$

$$I_A = (I_{rod})_A + (I_{disc})_A = \frac{0.32Wr^2}{g} + \frac{12.06Wr^2}{g} = \frac{12.38Wr^2}{g}$$

If i_A is the radius of gyration of the pendulum with respect to the perpendicular axis through A, we can write, $I_A = \frac{W + \frac{W}{6}}{g} i_A^2$ where $W + \frac{W}{6}$ is the total weight of the pendulum.

$$i.e., \quad \frac{W + \frac{W}{6}}{g} i_A^2 = \frac{12.38Wr^2}{g} \quad \gg\gg\gg \quad i_A^2 = 10.61 r^2$$

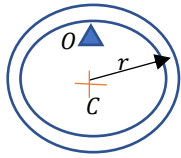
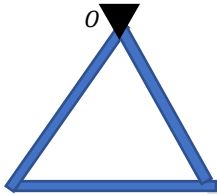
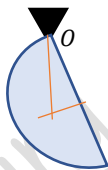
The distance of c.g from the axis of rotation,

$$c = \frac{1.2r \times \frac{W}{6} + W \times 3.4r}{W + \frac{W}{6}} = 3.09r$$

$$L_e = \frac{i_A^2}{c} = \frac{10.61 r^2}{3.09r} = 3.43r$$

$$T = 2\pi \sqrt{\frac{L_e}{g}} = 2\pi \sqrt{\frac{3.43r}{g}} = 11.65 \sqrt{\frac{r}{g}}$$

#Exercise 42

42.1 A thin circular hoop of mass per unit length m and mean radius r hangs over a pin at O at its circumference as shown. Find the period of small oscillations of the hoop in the vertical plane.	
42.2 A slender wire 90 cm long is bent in to the shape of an equilateral triangle and hangs from a pin at O as shown. Determine the period of small oscillations of the pendulum in the vertical plane.	
42.3 A homogeneous semicircular disc of radius r and weight W is suspended as a compound pendulum as shown and swings with small amplitude in the vertical plane. Calculate its period of oscillations.	

Prof. Biju N, School of Engineering, CUSAT